

KYZYL, Russian Federation

WMO: 360960

Lat: 51.717N

Lon: 94.500E

Elev: 632

StdP: 93.96

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.2	-38.3	-42.6	0.1	-40.2	-40.7	0.1	-38.3	3.4	-21.6	2.7	-23.7	0.4	60	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	32.7	17.3	30.5	16.8	28.5	16.2	19.1	27.9	18.2	26.9	17.4	25.7	1.9	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	12.6	21.2	15.3	11.7	20.7	14.3	11.0	20.5	57.2	27.8	54.2	26.8	51.6	26.0	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.1	4.1	-41.0	36.3	2.6	1.7	-42.8	37.5	-44.3	38.5	-45.8	39.5	-47.7	40.7		
(5)			-41.0	20.4	2.5	1.1	-42.9	21.1	-44.3	21.8	-45.8	22.4	-47.6	23.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.4	-28.7	-22.8	-8.5	5.9	12.1	19.3	21.3	18.2	11.1	2.1	-11.5	-24.9	(6)	
(7)		DBStd	18.19	5.39	6.50	7.57	5.16	4.55	3.93	3.36	3.56	4.16	4.42	6.86	6.38	(7)	
(8)		HDD10.0	4872	1201	919	572	139	31	0	0	0	35	247	644	1083	(8)	
(9)		HDD18.3	7065	1459	1153	831	372	199	36	11	46	218	504	894	1342	(9)	
(10)		CDD10.0	1066	0	0	0	17	95	279	349	255	69	2	0	0	(10)	
(11)		CDD18.3	217	0	0	0	0	6	64	102	43	2	0	0	0	(11)	
(12)		CDH23.3	2416	0	0	0	7	144	766	997	457	46	0	0	0	(12)	
(13)		CDH26.7	882	0	0	0	0	34	287	391	161	9	0	0	0	(13)	
(14)	Wind	WSAvg	1.4	0.7	0.9	1.4	2.1	2.3	2.0	1.7	1.6	1.6	1.3	1.0	0.7	(14)	
(15)	Precipitation	PrecAvg	258	9	5	6	9	16	38	54	52	31	11	14	13	(15)	
(16)		PrecMax	334	16	12	15	18	38	66	116	85	76	30	33	22	(16)	
(17)		PrecMin	199	4	2	2	1	5	10	20	17	11	2	3	5	(17)	
(18)		PrecStd	43	3	2	3	5	7	16	27	19	15	7	7	4	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-11.3	-5.3	10.8	24.1	30.2	34.7	36.0	34.4	27.9	18.0	5.4	-6.4	(19)	
(20)			MCWB	-11.9	-7.1	4.3	10.6	14.5	17.4	18.7	17.6	15.3	9.0	1.6	-7.3	(20)	
(21)		2%	DB	-15.8	-8.4	7.9	20.6	27.1	32.1	33.2	30.6	24.0	14.2	2.6	-10.7	(21)	
(22)			MCWB	-16.4	-9.6	2.8	9.3	12.9	16.5	18.2	17.3	13.7	7.2	-0.1	-11.3	(22)	
(23)		5%	DB	-18.4	-10.9	5.3	17.9	24.5	29.9	30.7	28.0	21.3	11.5	0.1	-13.0	(23)	
(24)			MCWB	-18.9	-11.9	1.1	7.8	11.7	16.1	17.5	16.8	12.6	5.8	-1.8	-13.5	(24)	
(25)		10%	DB	-20.6	-13.1	2.2	15.1	21.6	27.8	28.6	25.6	18.9	9.1	-2.1	-15.9	(25)	
(26)			MCWB	-21.0	-13.9	-0.7	6.7	10.4	15.3	16.8	15.8	11.5	4.3	-3.6	-16.3	(26)	
(27)		Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-11.8	-6.8	4.7	11.6	15.2	19.1	20.8	19.4	16.1	9.4	2.4	-7.6	(27)
(28)				MCDWB	-11.1	-5.4	9.7	22.0	28.0	29.2	30.6	27.5	25.4	16.8	4.5	-6.4	(28)
(29)	2%		WB	-16.4	-9.7	3.1	9.7	13.6	17.8	19.6	18.4	14.6	7.6	0.1	-11.3	(29)	
(30)			MCDWB	-15.8	-8.5	7.6	19.0	25.3	28.0	28.1	26.9	22.5	13.6	2.3	-10.7	(30)	
(31)	5%		WB	-18.8	-11.8	1.3	8.4	12.3	17.1	18.7	17.5	13.3	6.1	-1.7	-13.6	(31)	
(32)			MCDWB	-18.4	-10.9	4.7	16.6	22.5	26.5	27.3	25.7	20.1	11.0	0.0	-13.0	(32)	
(33)	10%		WB	-21.0	-13.9	-0.6	7.1	11.0	16.3	17.9	16.6	12.0	4.7	-3.6	-16.3	(33)	
(34)			MCDWB	-20.6	-13.1	2.2	14.6	20.3	25.7	26.2	24.2	17.9	8.5	-2.1	-15.9	(34)	
(35)	Mean Daily Temperature Range		5% DB	MDBR	8.1	11.5	12.3	12.4	12.9	12.8	11.5	11.7	12.5	11.3	8.2	6.9	(35)
(36)				MCDBR	8.5	11.7	12.3	17.1	18.4	16.4	15.2	16.1	16.3	15.3	9.7	6.9	(36)
(37)		MCWBR		8.1	10.6	7.7	8.3	7.6	5.7	4.9	5.9	7.4	8.8	7.1	6.4	(37)	
(38)		5% WB	MCDBR	8.5	11.8	11.9	15.2	16.3	13.6	12.0	13.3	14.4	14.6	9.1	6.9	(38)	
(39)			MCWBR	8.0	10.6	7.5	7.8	7.1	5.0	4.5	5.3	6.6	8.6	6.7	6.4	(39)	
(40)		Clear-Sky Solar Irradiance	taub	0.265	0.295	0.341	0.381	0.395	0.389	0.389	0.375	0.336	0.320	0.297	0.271	(40)	
(41)	taud		2.128	2.083	2.067	2.202	2.223	2.345	2.377	2.384	2.435	2.375	2.210	2.102	(41)		
(42)	Ebn at Noon		729	808	838	852	857	865	859	853	851	786	681	641	(42)		
(43)	Edh at Noon		73	103	129	129	132	118	113	106	90	78	68	64	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1.34	2.48	4.13	5.01	5.67	6.29	5.76	4.96	3.79	2.35	1.27	0.97	(44)		
(45)		RadStd	0.06	0.10	0.15	0.24	0.36	0.46	0.34	0.30	0.35	0.16	0.10	0.05	(45)		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page